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Automotive Heads-Up Display System

Group 2017.009

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1 High Level Description

1.1 Motivation

According to National Highway Traffic Association, 16% of fatalities and 20% of injuries related to car accidents in the United States are caused due to the driver's negligence [1]. One of the main causes of negligent driving is drivers diverting their attention from the road in order to check their phone, dashboard or the vehicle's infotainment system. If drivers can avoid looking away from the road to get the information they are looking for, the frequency of vehicular collisions can be decreased, making roads safer for all road users.

1.2 Project Objective

The objective of this project is to design a heads-up-display (HUD) system for automotive application. The HUD reduces/removes the need for the driver to look away from the roadway by projecting information desired by the user directly onto the vehicle's windshield.

1.3 Block Diagram

The high level architecture of the HUD system is shown below in **Figure 1,** and is divided into three subsystems: microcontroller, smartphone application and power.

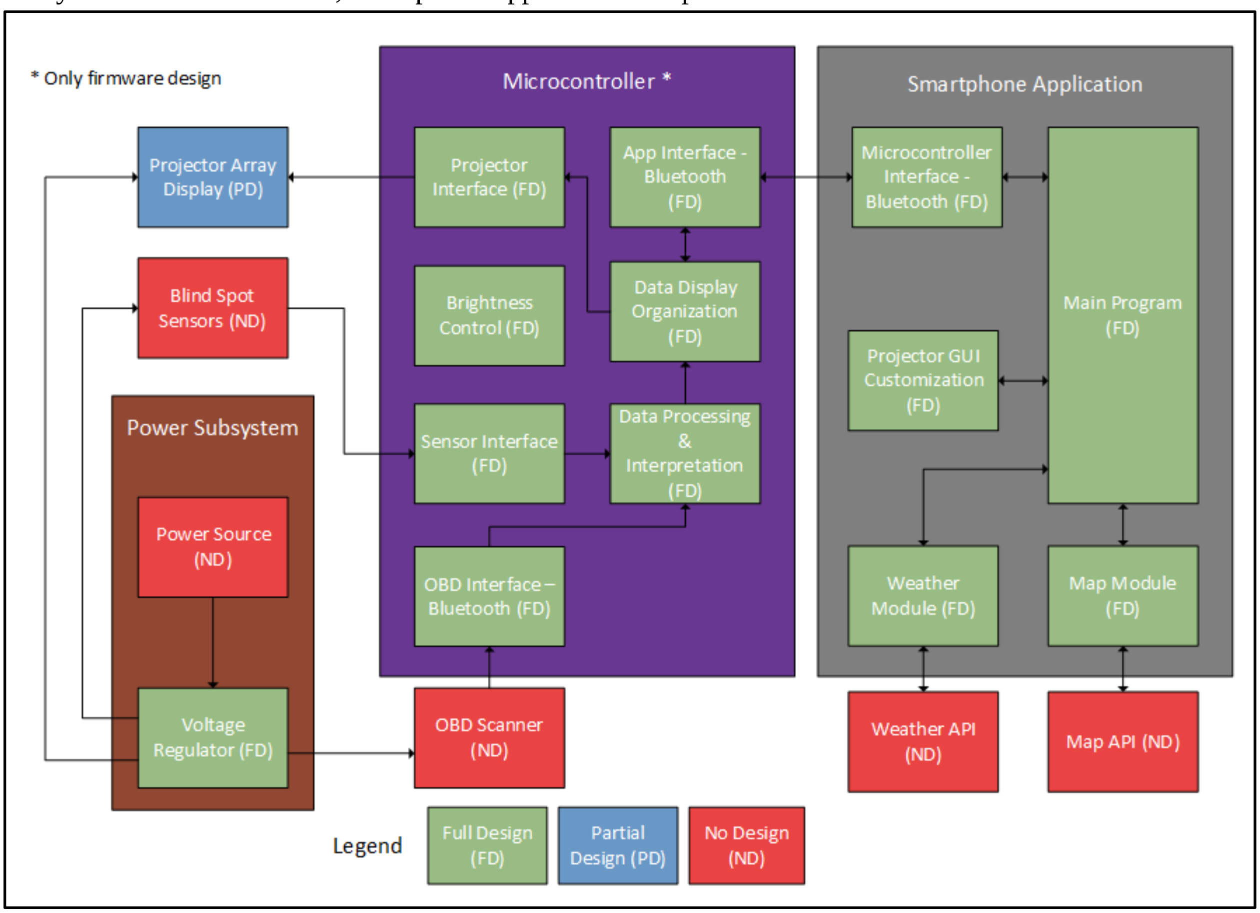

Figure 1: High Level Block Diagram

1.3.1 Microcontroller

The microcontroller gathers required data from sensors, scanners, and smartphones and also processes the data to display onto the projector arrays. Communication between the microcontroller and the smartphone app is done through a standard Bluetooth protocol stack. Each auxiliary device has an interface the microcontroller uses to communicate with the auxiliary device. Data from scanners and sensors are processed and interpreted before being displayed on the projectors. All of the microcontroller subsystems are designed and developed.

1.3.2 Smartphone Application

The smartphone application gathers data from a weather API (application-programmer interface) as well as a map API. It also sends the data to the microcontroller using a Bluetooth connection. The projector GUI (graphic user interface) customization can also be done on the smartphone. When the data is saved, the application will send the data to the microcontroller to change what is displayed over the projectors, their size, and location.

1.3.3 Power Subsystem

The power system comprises a power source and a voltage regulator to provide consistent, reliable power from the car. These components are based on existing integrated circuits, but the selection of these components based on their specifications and implementation will require design insight.

2 Project Specifications

Project specifications are classified into functional and non-functional specifications. Each of the specifications are further categorized into essential or non-essential, based on the importance of the specifications, in order to meet our project objective.

2.1 Functional Specifications

The functional specifications are design specifications that describe the operation of our automotive heads-up display system. **Table 1** below, lists the ten functional specifications.

2.2 Non-functional Specifications

Table 2 lists four non-essential functional specifications which describe the characteristics of our HUD device.

Table 1: Functional Specifications

Specification	Description	Classification
Maximum display size	The maximum display size must be at least 3.5 ft by 2.5 ft.	Essential
Minimum display size	The minimum display size must be within 1ft by 1ft.	Essential
Display on windshield	Display projected on windshield must be clearly readable to at least 9/10 people in sunny daytime conditions.	Essential
Adjust brightness	User must be able to adjust brightness of the heads-up-display (HUD).	Essential
Customize display elements	Users must be able to customize the location, size, and content of the display elements on the HUD.	Essential
GPS/Map display update rate	GPS/Map display must be continuously updated every 2 seconds or less.	Essential
Speed display update rate	Speed display must be continuously updated every 0.5 seconds or less.	Non-essential
Exceed speed limit warning	User must be notified if local speed limit has been exceeded within 0.5 seconds or less (synchronized with speed display update).	Non-essential
Weather warning alert update rate	Weather warnings/alerts must be continuously updated every 8 minutes or less.	Essential
Fuel display update rate	Fuel display must be continuously updated every 5 minutes or less.	Non-essential
Blind spot alert update rate	An alert to notify users of obstacles in the vehicle’s blind spot must be displayed within 0.1 seconds or less of the obstacle’s appearance.	Non-essential
Blind spot sensor detection size	Blind spot alert system must detect objects 5ft by 4ft or larger 100% of the time.	Non-essential

Table 2: Non-Functional Specifications

Specification	Description	Classification
System Volume	Total physical system volume must be 0.027 m^3 or less.	Essential
System Mass	Total physical system mass must be 2kg or less.	Non-essential
Power Consumption	Total system power consumption must be 30W or less.	Essential
System Compatibility	Heads-up-display system must be usable in any vehicle.	Non-essential

3 Risk Assessment

The major risks identified are detailed in **Table 3** which include technical limitations, material acquisition and group dynamics.

Table 3: Risk impact and probability identification with possible solutions to mitigate the risk

Situation	Nature of Situation	Probability	Impact
Technical Limitations	Materials required for design to meet specifications cannot be found/purchased. The success of this project depends heavily on the ability of the driver to read information presented on the HUD clearly under most normal weather conditions. This may be difficult to achieve without using advanced projection systems or specific optical films for the vehicle windshield that make the display visible under different lighting conditions. Additionally, there may be compatibility issues when various components of the HUD system are integrated together. To reduce the likelihood of this scenario, a significant amount of time will be spent during the prototyping stage to find potential equivalent replacements for components. Each component should be tested to ensure it meets the design specifications before integrating it into the overall design. This ensures incompatibility issues are detected early in the design process.	Medium	High
Material Acquisition	This project requires hardware components including, but not limited to mini projectors, sensors, integrated	Medium	High

	circuits, power adapters and optical films for the windshield. Materials ordered are subject to delays, damage-on-arrival and other logistical acquisition issues. To reduce the likelihood of this risk, materials required for the project should be obtained from reliable domestic sources. If a component is only available internationally, it must be ordered early in the prototype construction stage to account for potential delays. One team member should be responsible for organizing the billing and delivery of materials. Group members should check all materials to ensure there are no defective components. If defective components are found, replacements should be ordered without delay.		
Group Dynamics	Group's ability to work cooperatively and technical ability of individual members may not be as strong as expected. This can lead to significant delays in project timeline. In addition, it may be difficult to schedule group meetings due to conflicting schedules and availability of group members. To reduce the likelihood of these risks, group members should maintain a group calendar and schedule team meetings at least twice a week. This will help the team keep track of important deadlines and work together more efficiently. To minimize roadblocks due to lack of sufficient technical skills, team members should meet with the consultant regularly and spend some time each week to familiarize themselves with the tools and skills required for completion of the project.	Low	High

References

[1] *Road Safety in Canada - Transport Canada*, [Online]. Available: http://www.tc.gc.ca/eng/motorvehiclesafety/tp-tp15145-1201.htm#s36.